

Safety assessment of the substance (2E,2'E)-2,4,8,10-tetraoxaspiro[5.5]undecane-3,9-diylbis(2-methylpropane-2,1-diyl) bis(2-cyano-3-(3,4-dimethoxyphenyl)acrylate for use in plastic food contact materials

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The declarations of interest of all scientific experts active in EFSA's work are available at <https://open.efsa.europa.eu/experts>.

Abstract

The EFSA Panel on Food Contact Materials (FCM) assessed the safety of (2E,2'E)-2,4,8,10-tetraoxaspiro[5.5]undecane-3,9-diylbis(2-methylpropane-2,1-diyl) bis(2-cyano-3-(3,4-dimethoxyphenyl)acrylate, which is intended to be used as an additive at up to 0.75% w/w in polyethylene terephthalate (PET) single-use articles for contact with all types of food, for long-term storage at room temperature including hot fill and/or heating up to 100°C for a maximum of 2 h. Migration tests were carried out into the food simulants 3% acetic acid, 10% ethanol and olive oil, using PET samples containing the substance at the maximum intended level. The specific migration of the substance was up to 33 µg/kg food. The individual migration of its impurities was generally below the limits of quantification or not detected at the limits of detection, both in the few µg/kg food range, except for 3,4-dimethoxybenzaldehyde and spiroglycol in 10% ethanol (up to 45 and 46 µg/kg food, respectively). The specific migration of potential reaction or degradation products was below 50 µg/kg food. Genotoxicity studies demonstrated that the substance does not raise a concern. Impurities and the potential degradation or reaction products do not raise a concern for genotoxicity, based on genotoxicity studies, literature data or in silico assessment. The Panel concluded that the substance is not of safety concern for the consumers (i) if it is used as an additive at up to 0.75% w/w in PET intended for contact with all types of foods, except human milk, infant formula and water that could be used to reconstitute infant formula; (ii) if it is used for long-term storage at room temperature, including hot fill and/or heating conditions up to 100°C for a maximum of 2 h; and (iii) if the migration of its impurity 3,4-dimethoxybenzaldehyde does not exceed 50 µg/kg food.

KEYWORDS

2-cyano-3-(3,4-dimethoxyphenyl)-1,1'-[2,4,8,10-tetraoxaspiro[5.5]undecane-3,9-diyl bis(2,2-dimethyl-2,1-ethanediyl)]ester(2E,2'E), 2-propenoic acid, CAS No. 2233585-18-5, food contact materials, safety assessment

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1 | INTRODUCTION

1.1 | Background and Terms of Reference as provided by the requestor

Before a substance is authorised to be used in food contact materials (FCM) and is included in a positive list EFSA's opinion on its safety is required. This procedure has been established in Articles 8, 9 and 10 of Regulation (EC) No 1935/2004¹ of the European Parliament and of the Council of 27 October 2004 on materials and articles intended to come into contact with food.

According to this procedure, the industry submits applications to the Member States' competent authorities which transmit the applications to the European Food Safety Authority (EFSA) for their evaluation.

In this case, EFSA received an application from the German Federal Office of Consumer Protection and Food Safety, requesting the evaluation of the substance 2-propenoic acid,2-cyano-3-(3,4-dimethoxyphenyl)-,1,1'-[2,4,8,10-tetraoxaspiro[5.5]undecane-3,9-diylbis(2,2-dimethyl-2,1-ethanediyl)] ester, (2E,2'E), with the CAS number 2233585-18-5. The dossier was submitted on behalf of Chitec Technology Co., Ltd.

According to Regulation (EC) No 1935/2004 of the European Parliament and of the Council on materials and articles intended to come into contact with food, EFSA is asked to carry out an assessment of the risks related to the intended use of the substance and to deliver a scientific opinion.

2 | DATA AND METHODOLOGIES

2.1 | Data

The applicant has submitted a confidential and a non-confidential version of the dossier in support of their application for the authorisation of '2-Propenoic acid, 2-cyano-3-(3,4-dimethoxyphenyl)-, 1,1'-[2,4,8,10-tetraoxaspiro[5.5]undecane-3,9-diylbis(2,2-dimethyl-2,1-ethanediyl)] ester, (2E,2'E)' to be used in plastic FCM.

Additional information was provided by the applicant during the assessment process in response to a request from EFSA sent on 28 January 2025 (see Section 6).

In accordance with Art. 38 of the Commission Regulation (EC) No 178/2002² and taking into account the protection of confidential information and of personal data in accordance with Articles 39 to 39e of the same Regulation and of the Decision of the EFSA's Executive Director laying down practical arrangements concerning transparency and confidentiality,³ the non-confidential version of the dossier is published on Open.EFSA.⁴

According to Art. 32c(2) of Regulation (EC) No 178/2002 and to the Decision of EFSA's Executive Director laying down the practical arrangements on pre-submission phase and public consultations,⁵ EFSA carried out a public consultation on the non-confidential version of the application dossier from 16 September to 7 October 2025, for which no comments were received.

Data submitted and used for the evaluation are:

Non-toxicological data and information

- Chemical identity
- Description of manufacturing process of substance/FCM
- Physical and chemical properties
- Intended use
- Stability in food simulants
- Migration of the substance
- Identification, quantification and migration of reaction products and impurities

Toxicological data

- Bacterial gene mutation test and in vitro micronucleus test on the substance
- In vivo micronucleus tests and literature data on 3,4-dimethoxybenzaldehyde
- In silico and literature data on impurities

¹Regulation (EC) No 1935/2004 of the European parliament and of the council of 27 October 2004 on materials and articles intended to come into contact with food and repealing Directives 80/590/EEC and 89/109/EEC. OJ L 338, 13.11.2004, pp. 4–17.

²Regulation (EC) No 178/2002 of the European Parliament and of the Council of 28 January 2002 laying down the general principles and requirements of food law, establishing the European Food Safety Authority and laying down procedures in matters of food safety. OJ L 31, 1.2.2002, pp. 1–48.

³Decision available at <https://www.efsa.europa.eu/en/corporate-pubs/transparency-regulation-practical-arrangements>.

⁴The non-confidential version of the dossier, following EFSA's assessment of the applicant's confidentiality requests, is published on Open.EFSA and is available at the following link: <https://open.efsa.europa.eu/dossier/FCM-2024-30092>.

⁵Decision available at https://www.efsa.europa.eu/sites/default/files/corporate_publications/files/210111-PAs-pre-submission-phase-and-public-consultations.pdf.

2.2 | Methodologies

The assessment was conducted in line with the principles laid down in Regulation (EC) No 1935/2004 on materials and articles intended to come into contact with food. This Regulation requires applicants to submit an application accompanied by a technical dossier containing the information specified in the guidelines for the safety assessment of a substance to be published by EFSA. In practice, the technical dossier should contain the information required in the EFSA Note for Guidance for the preparation of an application for the safety assessment of a substance to be used in food contact materials (EFSA, 2008) and relevant cross-cutting documents from the EFSA Scientific Committee. The dossier submitted by the applicant was in line with these guidance documents.

The methodology is based on the characterisation of the substance that is the subject of the request for safety assessment prior to authorisation, its impurities and reaction and degradation products, the evaluation of the exposure to those substances through migration and the definition of minimum sets of toxicity data required for safety assessment.

To establish the safety from ingestion of migrating substances, the toxicological data indicating the potential hazard and the likely human exposure data need to be combined. Exposure is estimated from studies on migration into food or food simulants and considering that a person may consume daily up to 1 kg of food in contact with the relevant FCM.

As a general rule, the greater the exposure through migration, the more toxicological data is required for the safety assessment of a substance. Currently there are three tiers with different thresholds triggering the need for more toxicological information as follows:

- In case of high migration (i.e. 5–60 mg/kg food), an extensive data set is needed.
- In case of migration between 0.05 and 5 mg/kg food, a reduced data set may suffice.
- In case of low migration (i.e. <0.05 mg/kg food), only a limited data set is needed.

More detailed information on the required data is available in the EFSA Note for Guidance (EFSA, 2008).

The assessment was conducted in line with the principles described in the EFSA Guidance on transparency in the scientific aspects of risk assessment (EFSA, 2009) and considering the relevant cross-cutting guidance documents from the EFSA Scientific Committee.

3 | ASSESSMENT

According to the applicant, the substance '2-propenoic acid, 2-cyano-3-(3,4-dimethoxyphenyl)-,1,1'-[2,4,8,10-tetraoxaspiro[5.5]undecane-3,9-diylbis(2,2-dimethyl-2,1-ethanediyl)] ester, (2E,2'E)' with CAS No. 2233585-18-5, renamed by the Panel as '(2E,2'E)-2,4,8,10-tetraoxaspiro[5.5]undecane-3,9-diylbis(2-methylpropane-2,1-diyl) bis(2-cyano-3-(3,4-dimethoxyphenyl) acrylate' is an additive to be used as a UV light absorber in polyethylene terephthalate (PET) single-use articles. It is intended to be used at up to 0.75% w/w for contact with all types of food and long-term storage at room temperature, including hot fill and/or heating conditions up to 100°C for a maximum of 2 h. The applicant specified water bottles, beverage bottles and trays as typical articles in which the additive may be used. The applicant excluded the use in articles that may contact human milk and infant formula.

The substance has not been evaluated in the past by the Scientific Committee on Food (SCF) or EFSA.

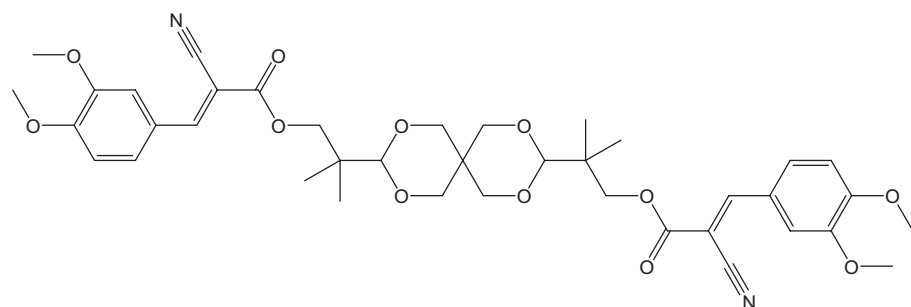
3.1 | Non-toxicological data⁶

3.1.1 | Identity of the substance

Chemical formula: C₃₉H₄₆N₂O₁₂

Molecular weight: 734.8 Da

Chemical structure:



⁶Technical dossier, Section 'Identity of substance'.

The substance is produced in a two-step process. The first step is [REDACTED] is reacted with 3,4-dimethoxybenzaldehyde (BZA), using [REDACTED] as a catalyst, to obtain [REDACTED] the second step is [REDACTED] reaction between [REDACTED] using [REDACTED] as a catalyst. Finally, the substance is purified [REDACTED] as solvents.

The applicant investigated the composition of the substance by targeted and non-targeted screening via high-performance liquid chromatography–quadrupole time of flight mass spectrometry (HPLC–Q-ToF–MS), HPLC–diode array detection (HPLC-DAD, 360 nm), gas chromatography–mass spectrometry (GC–MS) and inductively coupled plasma optical emission spectrometry (ICP-OES).

The applicant addressed a total of 14 compounds as potential impurities of the substance (Table 1; Appendix A). They were the [REDACTED] and solvents (i.e. BZA, spiroglycol, 403TM, [REDACTED]), by-products of the manufacturing process (2-cyano-3-(3,4-dimethoxyphenyl) acrylic acid, called ‘Acid’; the [REDACTED] product of [REDACTED], called ‘Single’), hydrolysis products of the catalyst [REDACTED] and five tentatively identified compounds (compounds A to E). Compounds A to E are structurally related to 403TM (for A and B), to Single (for C) and to the substance itself (for D and E).

Spiroglycol is authorised in Regulation (EU) No 10/2011⁷ as FCM No. 1052 with a specific migration limit (SML) of 5 mg/kg food. The catalyst [REDACTED] is expected to be removed during the purification step of the manufacturing process and was not considered by the applicant. Upon contact with moisture or water, the catalyst [REDACTED] is expected to rapidly degrade into [REDACTED] and [REDACTED], which are authorised in Regulation (EU) No 10/2011 [REDACTED]. The Panel noted that a fraction of [REDACTED] may be in the form of nanoparticles, which would not be covered by the authorisation. However, [REDACTED] is expected to be removed by the [REDACTED] process and it was not detected in the substance using a method with a limit of detection (LOD) of 0.0002% w/w. Furthermore, migration of embedded particles from PET into food is not expected to occur (Störmer et al., 2017). Consequently, the Panel did not further address [REDACTED] in this evaluation.

The purity of the substance was reported as >99.3%, calculated by subtracting the levels of the identified impurities from 100%.

TABLE 1 Compounds addressed as impurities in the substance.

	Name of the impurity	Content in the substance (% w/w)	Regulatory status (Regulation (EU) No 10/2011)
1	Acid	<0.00008 ^a	–
2	Spiroglycol	<0.008 ^b	FCM No. 1052, SML = 5 mg/kg
3	403TM	0.002	–
4	Single	0.097	–
5	BZA	<0.002 ^b	–
6	[REDACTED]	0.061	–
7	[REDACTED]	<0.04 ^b	–
8	Compound A	0.0004	–
9	Compound B	0.001	–
10	Compound C	0.005	–
11	Compound D	0.007	–
12	Compound E	0.009	–
13	[REDACTED]	<0.0002 ^b	[REDACTED]
14	[REDACTED]	<0.0001 ^b	[REDACTED]

^aBelow this limit of quantification (LOQ).
^bNot detected at this LOD.

3.1.2 | Physical and chemical properties⁸

3.1.2.1 | Physical properties

At room temperature, the substance is a yellow powder. According to the applicant, its melting and boiling points are 200–204°C and >380°C, respectively. The thermal stability of the substance was investigated via thermogravimetric analysis,

⁷Commission Regulation (EU) No 10/2011 of 14 January 2011 on plastic materials and articles intended to come into contact with food Text with EEA relevance. OJ L 12, 15.1.2011.
⁸Technical dossier, section ‘Physical and chemical properties’.

showing a loss of mass starting at around 300°C, which is higher than the maximum processing temperature of PET (280°C), with losses > 10% above 372°C. Therefore, the Panel concluded that the additive is thermally stable during the intended use.

The solubility of the substance in water was reported to be below 0.025 mg/L at 20°C, the octanol–water partition coefficient ($\log P_{ow}$) as 6.3.

The substance being a powder, the presence of a fraction of small particles should be established in accordance with the Guidance on technical requirements for regulated food and feed product applications to establish the presence of small particles, including nanoparticles (EFSA Scientific Committee, 2021). Since the melt temperature for making PET materials and articles by, e.g. injection moulding or extrusion is usually in the region of 240–270°C, the substance will melt and dissolve into the PET during the formulation and subsequent processing of the plastic. Therefore, any particle characteristics will be lost. Therefore, no additional assessment for the fraction of small particles was needed (EFSA Scientific Committee, 2021).

3.1.2.2 | Chemical properties

The applicant investigated the stability of the substance, Single and spiroglycol in food simulants. They were dissolved in 3% acetic acid or 10% ethanol and stored for 10 days at 60°C; then the food simulants were analysed via GC–MS and HPLC–Q-ToF–MS. The applicant tentatively identified various potential hydrolysis/degradation products and proposed hydrolysis pathways. The Panel noted that the study had severe limitations, such as a reported concentration of the substance that was many orders of magnitude higher than its solubility, and the identification of compounds that are unlikely to have been generated via hydrolysis. Therefore, the Panel did not further consider the study in this evaluation. However, the Panel considered it plausible that the substance hydrolyses during migration testing, in contact with food and under gastrointestinal conditions. Therefore, the Panel identified the potential hydrolysis products considering two potential hydrolysis pathways in acidic environment: hydrolysis of the acetal and the ester bonds. Hydrolysis of the acetal bonds of the spiroglycol moiety is expected and was already addressed in the evaluation of spiroglycol as a starting substance (EFSA CEF Panel, 2014). The potential hydrolysis products (seven compounds) identified by the Panel are reported in Appendix A.

The applicant also investigated the potential formation of degradation products by GC–MS and HPLC–Q-ToF–MS after exposing the substance to simulated sunlight for 10 days. Eighteen compounds were reported as potential ultraviolet light (UV) degradation products. Eight of them were already identified as impurities of the substance (No. 1–6, 10 and 12 in Table 1). The other 10 are listed in Appendix A and were 3,4-dimethoxybenzoic acid, 3,4-dimethoxycinnamionitrile and eight tentatively identified compounds (UV-Unknowns A–H). The applicant did not quantify the identified compounds. The Panel considered that the proposed compounds are unlikely to be generated by UV irradiation, based on their chemical structure and considering that cyanoacrylate UV absorbers are reported to be stable against photodegradation (Pickett, 2000). Therefore, the origin of the proposed degradation products could not be elucidated. The Panel further assessed them in the evaluation considering their structural similarity with the substance and its impurities.

3.1.3 | Specific migration⁹

The migration of the substance and six impurities (No. 1–6, i.e. Acid, spiroglycol, 403TM, Single, BZA [REDACTED]) was investigated in migration tests using PET films with a thickness of 100 µm and a density of 1.35 g/cm³, produced using the substance at the maximum intended level. Tests were carried out at 60°C for 10 days using the food simulants 3% acetic acid, 10% ethanol and olive oil. The specific migration was determined by GC–MS and HPLC–tandem mass spectrometry (HPLC–MS/MS).

The migration of four other impurities (No. 7–9, 14, i.e. [REDACTED], compounds A and B, [REDACTED]) was calculated via total mass transfer considering a 100 µm-thick PET sample containing the substance at the maximum intended use level. The migration of the remaining three impurities (No. 10–12: compounds C, D, E) was determined via migration modelling.

The migration of the compounds proposed by the applicant as potential UV degradation products (Section 3.1.2.2) was investigated in migration tests using PET specimens that were exposed to simulated sunlight for 10 days. The testing conditions and analytical methods were the same as those used for the substance and its impurities.

An overview of the results is provided in Table 2.

⁹Technical dossier, section 'Data on migration of substance'.

TABLE 2 Overview of the migration results.

Name of the migrating compound	Specific migration (µg/kg food)		
	3% acetic acid	10% ethanol	Olive oil
Migration tests carried out at 60°C for 10 days			
Substance	<2 ^a	<2 ^a	33
Acid	<2 ^a	<2 ^a	<2 ^a
Spiroglycol	<2 ^a	46	<2 ^a
403TM	<2 ^a	<2 ^a	<2 ^a
Single	<2 ^a	<2 ^a	<2 ^a
BZA	<5 ^b	45	<2 ^a
██████████	<2 ^a	<2 ^a	<37 ^c
██████████	0.24 ^c	0.24 ^c	0.24 ^c
Compound A	0.23 ^c	0.23 ^c	0.23 ^c
Compound B	0.42 ^c	0.42 ^c	0.42 ^c
Compound C	0.053 ^d	0.053 ^d	0.053 ^d
Compound D	0.038 ^d	0.038 ^d	0.038 ^d
Compound E	0.045 ^d	0.045 ^d	0.045 ^d
██████████	0.6 ^e	0.6 ^c	0.6 ^c
Migration tests carried out at 60°C for 10 days under simulated sunlight			
3,4-Dimethoxy benzoic acid	<20 ^a	<50 ^e	<50 ^a
3,4-Dimethoxycinnamonnitrile	<50 ^a	<50 ^a	<20 ^a
UV-Unknowns A-H	<10 ^a (for each compound)	<10 ^a (for each compound)	<20 ^a (for each compound)

^aNot detected at this LOD.
^bBelow this LOQ.
^cCalculated via total mass transfer.
^dDetermined via migration modelling (MIGRATEST EXP software).
^eNot detected at the LOD, then corrected by the Panel based on the available raw data.

3.1.3.1 | Migration of the substance and its hydrolysis products

The specific migration of the substance in aqueous simulants was not detected at a LOD of 2 µg/kg food. However, as the reported recovery was extremely low (close to 0%), the actual migration of the substance into aqueous simulants may be higher than 2 µg/kg food. The specific migration in olive oil was up to 33 µg/kg food, with a recovery around 100%. Considering the lipophilicity of the substance (log Pow=6.3), the Panel considered that its potential maximum migration into aqueous simulants (i.e. not considering a potential hydrolysis) would be lower than 33 µg/kg food.

As the diffusion of the substance to the surface of PET is much slower than its hydrolysis in food, the Panel considered that the specific migration of each potential hydrolysis product (Section 3.1.2.2) and their sum will not exceed 33 µg/kg food.

The Panel noted that some of the potential hydrolysis products are also impurities (Single, spiroglycol, Acid). Considering the total mass transfer based on their content in the substance (Section 3.1.1), their migration would still be below 50 µg/kg food.

3.1.3.2 | Migration of impurities and potential reaction and degradation products

The individual specific migration of ██████████, compounds A to E and ██████████ was determined via total mass transfer or migration modelling and was up to 0.6 µg/kg food.

The specific migration of the other impurities (Acid, spiroglycol, 403TM, Single, BZA, ██████████) was measured via migration tests.

The recoveries for Acid, spiroglycol, 403TM and Single in aqueous simulants were extremely low (e.g. close to 0% for Single), except for spiroglycol in 10% ethanol (≈100%). Therefore, their actual migration into aqueous simulants may be higher than measured (i.e. not detected at a LOD of 2 µg/kg food, Table 2). However, it would be below 50 µg/kg food based on considerations on their total mass transfer and hydrolysis of the substance (Section 3.1.3.1). Since the recoveries in olive oil were close to 100%, the Panel considered the corresponding migration results as reliable. All four impurities were not detected at a LOD of 2 µg/kg food in olive oil, except for spiroglycol in 10% ethanol which migrated at up to 46 µg/kg food.

For the remaining impurities (BZA and ██████████), the migration results were considered reliable as supported by appropriate recovery values in all simulants. The migration of BZA was up to 45 µg/kg (10% ethanol), below the LOQ of 5 µg/kg food (3% acetic acid) and not detected at the LOD of 2 µg/kg food (olive oil). The migration of ██████████

was not detected at the LODs of 2 µg/kg food (aqueous simulants) and 100 µg/kg food (olive oil). However, as their total mass transfer based on the content in the substance (i.e. 0.061% w/w, Table 1) is 37 µg/kg food,¹⁰ the actual migration of in olive oil, if any, would be below 50 µg/kg food.

Each of the 10 compounds that were proposed as UV degradation products (3,4-dimethoxy benzoic acid, 3,4-dimethoxycinnamionitrile, UV-Unknowns A to H, Section 3.1.2.2) were not detected at LODs ranging from 10 to 50 µg/kg food. For 3,4-dimethoxybenzoic acid, the reported LOD in 10% ethanol was 100 µg/kg food, but the Panel calculated that it is below 50 µg/kg food based on the provided raw data.

The Panel concluded that the potential migration of the substance and each impurity, potential hydrolysis product and proposed UV degradation product is lower than 50 µg/kg food.

3.2 | Toxicological data¹¹

As the migration of the substance, the impurities and reaction products was below 50 µg/kg food, in accordance with the 'EFSA Note for Guidance' (EFSA, 2008), the applicant submitted only data to assess their genotoxic potential.

3.2.1 | Genotoxicity

The applicant provided a bacterial reverse mutation assay (Ames) test and an in vitro micronucleus test to address the genotoxicity potential of the substance. For the impurity BZA, the applicant submitted literature data and micronucleus tests (both in vitro and in vivo). The genotoxicity potential of the impurities migrating above 0.15 µg/kg food, i.e. the migration corresponding to the threshold of toxicological concern for genotoxic carcinogens (0.0025 µg/kg body weight per day, EFSA Scientific committee, 2012), was addressed via in silico evaluations and literature data.

3.2.1.1 | Genotoxicity of the substance

The bacterial reverse mutation assay was performed in compliance with USA Good Laboratory Practices (GLP) regulations, compatible with the Organisation for Economic Co-operation and Development (OECD) principles and according to the OECD TG 471 (OECD, 1997). Five bacteria test strains were used, namely *Salmonella Typhimurium* TA1535, TA1537, TA98, TA100 and *Escherichia coli* WP2 *uvrA*. The test substance was evaluated in two experiments: in a plate incorporation assay in the absence and presence of metabolic activation (rat liver S9 mix) and in a pre-incubation assay using a similar experimental design. Each bacteria strain was exposed to the substance (dissolved in dimethyl sulfoxide (DMSO)) at 1.58, 5.0, 15.8, 50, 158, 500, 1580 and 5000 µg/plate as well as negative (vehicle) or positive controls in triplicate plates. Precipitation was observed in both assays at ≥ 500 µg/plate, both with and without S9, except in the plate incorporation assay with *S. Typhimurium* TA1535 (≥ 1580 µg/plate); precipitation hiding the background lawn at ≥ 1580 µg/plate was noted. Toxicity (as indicated by a slight decrease in the number of revertant colonies) was found at the highest concentration for strain TA1537 in the pre-incubation method without S9 and for strain TA98 with both methods at doses ≥ 1580 µg/plate with and without S9. The results obtained for the negative and positive controls were within the respective historical control range for each strain used. In both assays, no increase in the number of revertant colonies over the control was observed at any of the concentrations tested in the absence or presence of metabolic activation, indicating that the substance did not induce gene mutations under the conditions tested. The Panel considered the study reliable without restrictions and the negative results of high relevance.

The in vitro micronucleus test was performed in a GLP test facility, using the cytokinesis-block method with human peripheral blood lymphocytes (PBL), according to the OECD TG 487 (OECD, 2016b). Short (3 h) and extended (24 h) exposures of PBL to the test substance were done in duplicate cultures. The short exposure test was conducted with and without metabolic activation (rat S9 fraction) and was followed by a 24 h recovery period in the presence of cytochalasin B; the extended exposure test was without metabolic activation in the presence of cytochalasin B. For both exposure periods, the selected concentration range of the test substance (dissolved in DMSO) comprised 125, 250, 500, 1000 and 2000 µg/mL; at 3 h exposure, 62.5 µg/mL was also tested. Appropriate negative (vehicle) and positive controls were used. Micronuclei were scored in 2000 cytokinesis-blocked (binucleated) cells per each concentration and test condition to determine the frequency of micronucleated cells. The cytokinesis-block proliferation index was calculated for each treatment condition as a measure of toxicity. There was no evidence of precipitation of the substance at any concentration tested. A positive response was observed for the positive controls. No increase in the frequencies of micronucleated binucleated cells and no toxicity were observed in treated cultures compared with the respective vehicle controls, irrespective of the exposure duration and absence or presence of metabolic activation.

The Panel considered the results of the test as negative, i.e. that under the conditions tested the test substance did not induce the formation of micronuclei. The study was considered reliable without restrictions and the negative results of high relevance.

¹⁰calculated for a 100 µm-thick PET sample containing the substance at the maximum intended level.

¹¹Technical dossier, Section 'Toxicological data'.

Overall, based on adequate negative data in the Ames and in vitro micronucleus tests, the Panel concluded that the substance does not raise concern for genotoxicity.

3.2.1.2 | Genotoxicity of impurities

Fourteen compounds were addressed as impurities (Table 1). The specific migration of three of them (compounds C, D and E) was estimated to be below 0.15 µg/kg food and, therefore, their genotoxicity potential was not addressed. The use of spiroglycol and [REDACTED] is authorised by Reg. (EU) No 10/2011 and their migration was compliant with the restrictions, i.e. a SML of 5 mg/kg food for spiroglycol (FCM No. 1052) and no SML for [REDACTED] (FCM No. [REDACTED]). As reported in Section 3.1.1, the migration of [REDACTED] is not expected.

The specific migration of the remaining eight impurities (BZA, Acid, 403TM, Single, [REDACTED], compounds A and B) was below 50 µg/kg food, but above 0.15 µg/kg food. Therefore, their genotoxicity potential was addressed in this section.

For BZA, the applicant provided literature data (bacterial reverse mutation assay and in vitro mammalian cell micronucleus assay) and unpublished genotoxicity studies (in vitro and in vivo micronucleus tests). Regarding the literature data, the mutagenic activity of BZA was addressed by Mortelmans et al. (1986), reporting a bacterial reverse mutation assay with *S. Typhimurium* strains (TA1535, TA1537, TA98 and TA100) in the absence and presence of metabolic activation according to OECD TG 471. BZA did not induce a significant dose-related increase in the frequency of revertants, indicating a lack of mutagenicity under the tested conditions. Thakkar et al. (2022) reported a positive response in an in vitro mammalian cell micronucleus assay performed in accordance with OECD TG 487 and a negative result in an in vivo micronucleus test performed in compliance with GLP and following OECD TG 474 (OECD, 2014). The negative result of a follow-up micronucleus test in 3D reconstructed skin was reported, but, given the difference between oral exposure and this in vitro model that mimics dermal exposure, the Panel considered the study of low relevance for the present evaluation.

Regarding the unpublished genotoxicity studies, an in vitro cytokinesis-block micronucleus assay was conducted using human PBL, exposed to 1064, 1330 and 1662 µg/mL BZA for 3 h in the presence and absence of metabolic activation (S9) and at 211, 362 and 496 µg/mL for 24 h without S9. For the longer exposure, the top concentration was limited by cytotoxicity (approximately 57% for the highest concentration tested); for the shorter exposure, cytotoxicity was below 50%. Statistically significant and dose-dependent increases in the frequencies of micronucleated binucleated cells were observed in the 24 h treatment at 362 and 496 µg/mL, although only the highest concentration was outside the historical vehicle control range. No positive response was observed for the 3 h treatments. The frequencies of micronucleated cells of vehicle and positive controls were within the historical control range. In view of the reported results, the BZA was concluded to be positive for the induction of micronuclei in mammalian cells. The Panel considered the study reliable without restrictions and the positive results of high relevance.

An in vivo micronucleus test was carried out in compliance to GLP and following OECD TG 474 (OECD, 2014) in groups of Hsd:ICR (CD-1) male and female mice exposed to BZA via oral gavage on two consecutive days (24 h interval) at doses of 500, 1000 or 2000 mg/kg bw and euthanised at 48 h after the last administration. The frequency of peripheral blood reticulocytes (RET) with micronuclei was determined by flow cytometry of up to 20,000 RET per animal. The percentage of RET served as a measure of cytotoxicity of the test item in bone marrow and in blood. The test material did not induce a statistically significant increase in the incidence of RET containing micronuclei for none of the treated mice groups compared to the vehicle control group. No reduction of the percentage of RET was observed in the treated groups, indicating that BZA did not induce cytotoxicity in the bone marrow. The Panel noted that no evidence of toxicity to the bone marrow indicative of the exposure of the target tissue was provided. However, it noted the presence of clinical signs in treated animals, including central nervous system (CNS) depression, which are indicative of systemic exposure (EFSA Scientific Committee, 2017b). The Panel also noted an increase of the percentage of RET in males exposed to the highest doses, which might be interpreted as an indication of compensatory erythropoiesis, i.e. as indirect evidence of bone marrow exposure. In addition, BZA is a small and mildly lipophilic molecule that can reach the bone marrow via blood circulation. Based on these observations, the Panel considered that it is likely that BZA reached the bone marrow after oral treatment and, hence, the study can be considered reliable without restrictions.

Overall, the available information can be considered sufficient to exclude a concern for the genotoxicity of BZA.

The genotoxicity of the remaining impurities (Acid, 403TM, Single, [REDACTED], compounds A and B) was addressed by an in silico evaluation with quantitative structure–activity relationship (QSAR) models or literature data.

For Acid, 403TM and Single, the applicant carried out an in silico evaluation using the OECD QSAR Toolbox. The results for 403TM and Single were the same. For both impurities, predictions for general mechanistic profilers gave positive calls for DNA binding and protein binding as 'Michael addition'. However, the remaining general mechanistic predictions provided negative results or no values. Regarding predictions from endpoint-specific profilers, negative prediction calls were reported for 'in vitro mutagenicity (Ames test) alerts by ISS'¹² and 'DNA alerts for Ames, chromosomal aberrations and in vitro micronucleus test by OASIS'.¹³ Both impurities triggered the H-acceptor-path3-H-acceptor alert for the 'in vivo mutagenic-

¹²ISS: Istituto Superiore di Sanità.

¹³OASIS: database of the Laboratory of Mathematical Chemistry University, Burgas, Bulgaria.

ity (micronucleus) alerts by ISS'. However, this alert was shown to be devoid of predictive value (Aljallal et al., 2024; Pradeep et al., 2021) and, hence, disregarded. The predictions for Acid were the same as those for 403TM and Single, except for the OASIS profiler that gave an AN2 alert for 'Ames, chromosomal aberrations and in vitro micronucleus test', which, however, was not supported by the negative prediction of the ISS profiler for the 'in vitro mutagenicity (Ames test) alerts'. Finally, results from two other complementary silico models, providing expert-rule-based (Derek Nexus), and statistical-based (Sarah Nexus) predictions for mutagenicity and genotoxicity, predicted the three impurities to be negative for in vitro mutagenicity and devoid of structural alerts for genotoxicity. Overall, the Panel considered that Acid, 403TM and Single do not raise concern regarding their genotoxicity.

For [REDACTED], the applicant provided publications reporting negative results in the Ames test ([REDACTED]) and an in vivo mammalian erythrocyte micronucleus test ([REDACTED]) that indicated no concern regarding their genotoxicity.

For [REDACTED], an OECD report¹⁴ reported that under physiological conditions it is likely to be metabolised to [REDACTED]. [REDACTED] and [REDACTED] lack mutagenic potential, based on the negative result of an Ames test performed with and without metabolic activation. [REDACTED] did not show clastogenic activity, based on a negative cytogenetic assay in V79 Chinese Hamster lung cells, in the presence and absence of metabolic activation. Both tests were conducted following the respective OECD test guideline and in compliance with GLP. For both compounds, there were no reported structural alerts for genotoxicity. A QSAR analysis (OECD Toolbox 4.4) confirmed that [REDACTED] does not trigger structural alerts for genotoxicity by endpoint-specific profilers, including clastogenic and aneugenic effects. Overall, based on the available information, the Panel considered that there is no indication of a genotoxic potential for [REDACTED] and its metabolite [REDACTED].

For compounds A and B, which are structurally similar to 403TM (Section 3.1.1), the genotoxicity potential was addressed via the OECD QSAR Toolbox 4.4. Both compounds produced, in a general mechanistic profiler, an alert for Michael addition due to the presence of a benzene ring. The alert was not supported by predictions of endpoint-specific profilers, comparable to the findings reported for 403TM. Likewise, the alert triggered for in vivo genotoxicity by ISS 'H-acceptor-path3-H-acceptor' lacks predictive value. Overall, the Panel considered that compounds A and B did not raise any relevant alert regarding their genotoxicity.

3.2.1.3 | Genotoxicity of potential reaction and degradation products

As reported in Section 3.1.3, the Panel considered that the potential migration of the sum of the possible hydrolysis products of the substance (seven products, Appendix A) would be below 33 µg/kg food, i.e. below 50 µg/kg food. As a migration above 0.15 µg/kg food could not be ruled out for each hydrolysis product, their genotoxicity potential needed to be addressed. The safety of the hydrolysis products that are also impurities of the substance (spiroglycol, Acid, Single) was already addressed in Section 3.2.1.2. The safety of the hydrolysis products that result from the degradation of spiroglycol (hydroxy pivalaldehyde and pentaerythritol) was already considered in the 2014 evaluation of spiroglycol as a monomer (EFSA CEF Panel, 2014). For the remaining two hydrolysis products (A and B), an in silico evaluation performed with the OECD QSAR Toolbox 4.4 identified a potential for Michael addition due to the presence of a benzene ring, which was not supported by predictions of endpoint-specific profilers; the alert for in vivo genotoxicity (H-acceptor-path3-H-acceptor) lacks predictive value and was disregarded. Additionally, for hydrolysis products A the presence of an aldehyde group triggered an alert in the OECD QSAR toolbox for direct genotoxicity via reaction with the amine group of nucleobases and originated a positive prediction call for 'in vitro mutagenicity (Ames) by ISS'. However, negative prediction calls were reported for 'DNA alerts for Ames, CA and MN by OASIS' and for 'protein binding alerts for CA'. To confirm these findings, the Panel performed an additional in silico assessment applying a suite of QSAR statistical models (VEGA platform, version 1.2.3). VEGA models predicted the hydrolysis product A to be non-mutagenic in the Ames test and, with good reliability, inactive in the in vivo micronucleus test, overruling the positive predictions provided by models for in vitro activity. In view of these findings, the Panel considered that none of the hydrolysis products displays relevant genotoxic potential.

The specific migration of the 10 compounds that were proposed by the applicant as potential UV degradation products was not detected at LODs ranging from 10 to 50 µg/kg food. As a migration above 0.15 µg/kg food could not be ruled out, their genotoxicity potential was assessed using the OECD QSAR Toolbox 4.4. The Panel noted that the structures of the reported compounds bear a benzene ring with the potential to form reactive quinones with the DNA molecule. As such, all compounds raised an alert for a Michael addition, but the predictions of endpoint-specific profilers for mutagenicity, genotoxicity and carcinogenicity did not support it; the alert for in vivo genotoxicity (H-acceptor-path3-H-acceptor), which was raised for all compounds, lacks predictive value and was disregarded. Additionally, the compound 'UV-Unknown G' triggered the same predictions obtained for 'hydrolysis product A' due to the presence of an aldehyde group. However, the compound was predicted as non-mutagenic in the Ames test and inactive in the in vitro micronucleus test by in silico assessment via the VEGA platform. Overall, no genotoxic potential was identified for the compounds that were proposed by the applicant as potential UV degradation products.

¹⁴ [REDACTED]

4 | DISCUSSION

The substance (2E,2'E)-2,4,8,10-tetraoxaspiro[5.5]undecane-3,9-diylbis(2-methylpropane-2,1-diyl) bis(2-cyano-3-(3,4-dimethoxyphenyl)acrylate) is intended to be used as an additive in PET single-use articles. It is intended to be used at up to 0.75% w/w for contact with all types of food and long-term storage at room temperature, including hot fill and/or heating conditions up to 100°C for a maximum of 2 h. The contact with infant formula and human milk was excluded.

The specific migration of the substance, six of its impurities and the proposed UV degradation products was determined via migration tests into 3% acetic acid, 10% ethanol and olive oil. The Panel considered that the test conditions cover the intended uses, including contact with solid food. Due to the very low recovery, the specific migration into aqueous simulants could not be determined, but the Panel estimated that it would not exceed the migration measured in olive oil, i.e. 33 µg/kg food. The substance may be hydrolysed, especially under acidic conditions, but the migration of the sum of the potential hydrolysis products would not exceed 33 µg/kg food. The migration of the impurities was not detected at LODs or detected below LOQs in the low µg/kg food range, except for BZA and spiroglycol into 10% ethanol (up to 45 and 46 µg/kg food, respectively). The migration of the compounds proposed as potential UV degradation products was not detected at LODs below 50 µg/kg food. The migration of the remaining impurities was determined via total mass transfer or migration modelling and was below 50 µg/kg food.

Based on the results of an in vitro bacterial reverse mutation assay and an in vitro mammalian cell micronucleus test, the Panel concluded that the substance does not raise a concern for genotoxicity. Based on experimental data, the Panel considered that BZA does not raise a concern for genotoxicity. In silico and literature data on the impurities migrating above 0.15 µg/kg food indicated the absence of genotoxic potential. Based on an in silico assessment, the Panel considered the potential hydrolysis products and the compounds proposed by the applicant as potential UV degradation products of no concern for genotoxicity. Overall, the Panel concluded that the substance, its impurities and its potential hydrolysis and reaction products do not raise a concern for genotoxicity.

The specific migration of BZA should be limited to 50 µg/kg food, as only data on its genotoxicity potential was provided.

The substance is intended to be used in PET bottles in contact with water that could be used to reconstitute infant formula (such as from powdered formula). Therefore, the EFSA Guidance on the risk assessment of substances present in food intended for infants below 16 weeks of age (EFSA Scientific Committee, 2017a) was considered. The toxicological data submitted by the applicant supports a migration ≤ 3 µg/kg food (corresponding to Tier I, see Annex I of the guidance). The migration into water was not determined. As the specific migration in aqueous simulants was estimated to be below 33 µg/kg food, a potential migration in water above 3 µg/kg food cannot be ruled out. Therefore, the safety of the use in contact with water that could be used to prepare infant formula was not demonstrated.

5 | CONCLUSIONS

Based on the above-mentioned data, the FCM Panel concluded that the substance (2E,2'E)-2,4,8,10-tetraoxaspiro[5.5]undecane-3,9-diylbis(2-methylpropane-2,1-diyl) bis(2-cyano-3-(3,4-dimethoxyphenyl)acrylate) is not of safety concern for the consumers if:

- it is used as an additive to manufacture PET at up to 0.75% w/w for contact with all type of food, except human milk, liquid infant formula and the water that could be used to reconstitute infant formula; and
- it is used for long-term storage at room temperature, including hot fill and/or heating conditions up to 100°C for a maximum of 2 h; and
- the migration of its impurity 3,4-dimethoxybenzaldehyde does not exceed 50 µg/kg food.

6 | DOCUMENTATION AS PROVIDED TO EFSA

Dossier “2-propenoic acid, 2-cyano-3-(3,4-dimethoxyphenyl)-,1,1'-[2,4,8,10-tetraoxaspiro[5.5]undecane-3,9-diylbis(2,2-dimethyl-2,1-ethanediyl)] ester, (2E,2'E)". January 2025. Submitted by Chitec Technology Co., Ltd.

Additional data, October 2025. Submitted by Chitec Technology Co., Ltd.

ABBREVIATIONS

403TM	ethyl-(2Z)-2-cyano-3-(3,4-dimethoxyphenyl)prop-2-enoate
Acid	2-cyano-3-(3,4-dimethoxyphenyl) acrylic acid
BZA	3,4-dimethoxybenzaldehyde
CEF Panel	EFSA Panel on Food Contact Materials, Enzymes, Flavourings and Processing Aids
CNS	central nervous system
Da	Dalton
DMSO	dimethyl sulfoxide
FCM	food contact materials
GC–MS	gas chromatography–mass spectrometry

GLP	good laboratory practice
HPLC–DAD	high-performance liquid chromatography–diode array detection
HPLC–MS/MS	high-performance liquid chromatography–tandem mass spectrometry
HPLC–Q-ToF–MS	high-performance liquid chromatography–quadrupole time of flight mass spectrometry
ICP–OES	and inductively coupled plasma optical emission spectrometry
LOD	limit of detection
LOQ	limit of quantification
OECD	Organisation for Economic Co-operation and Development
PBL	human peripheral blood lymphocytes
PET	polyethylene terephthalate
logPo/w	octanol/water partition coefficient
QSAR	quantitative structure–activity relationship
RET	peripheral blood reticulocytes
SCF	Scientific Committee on Food
SML	specific migration limit
spiroglycol	β3,β3,β9,β9-tetramethyl-2,4,8,10-tetraoxaspiro[5.5]undecane-3,9-diethanol
UV	ultraviolet light

REQUESTOR

German Competent Authority

QUESTION NUMBER

EFSA-Q-2024-00604

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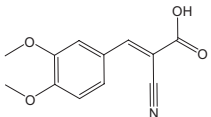
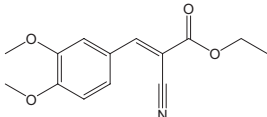
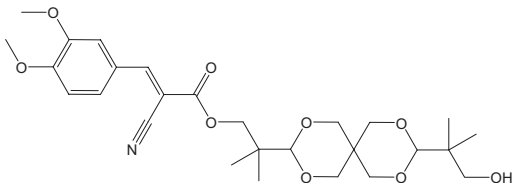
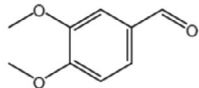
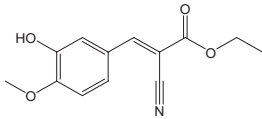
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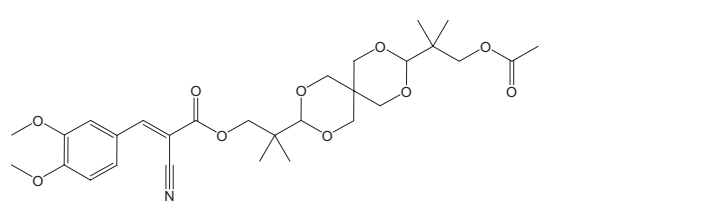
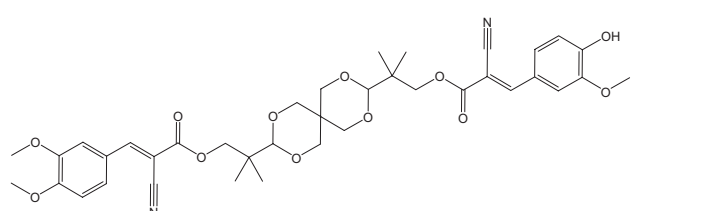
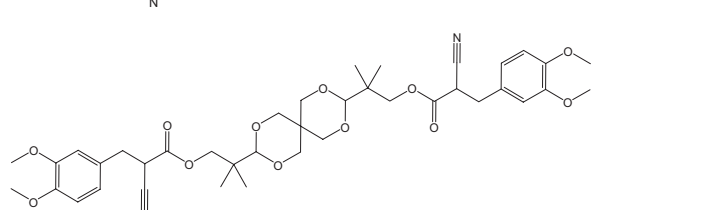
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APPENDIX A

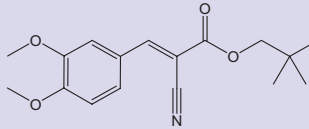
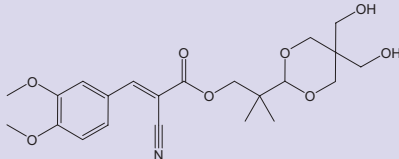
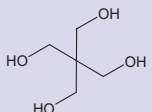
List of compounds considered in the assessment

Name of the compound	Structure
A.1 Impurities addressed in the compositional analysis (Section 3.1.1)	
Acid	
Spiroglycol	
403TM	
Single	
BZA	
Compound A	

(Continues)

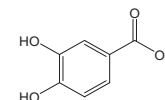
Name of the compound	Structure
Compound B	
Compound C	
Compound D	
Compound E	
	
	
A.2 Potential hydrolysis products considered by the Panel (Section 3.1.2.2)	
Single	See A1
Spiroglycol	See A1
Acid	See A1

(Continues)

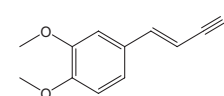
Name of the compound	Structure
Hydrolysis product A	
Hydrolysis product B	
Pentaerythritol	
Hydroxy pivalaldehyde	

A.3 Additional substances proposed by the applicant as UV degradation products (Section 3.1.2.2)

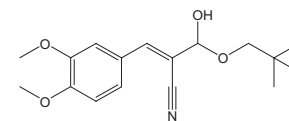
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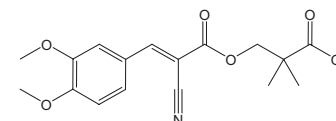
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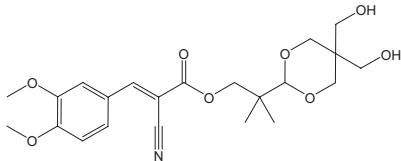
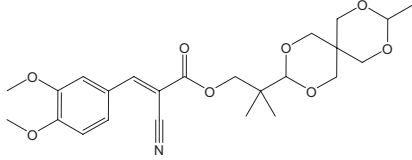
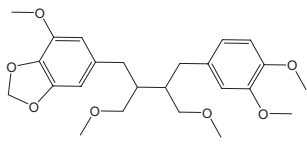
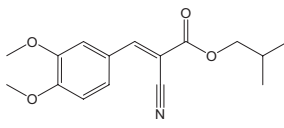
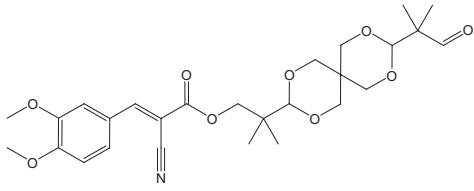
UV-Unknown A



UV-Unknown B



(Continues)

Name of the compound	Structure
UV-Unknown C	
UV-Unknown D	
UV-Unknown E	
UV -Unknown F	
UV-Unknown G	
UV-Unknown H	